

The value iteration method for countable state Markov decision processes

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Abstract

This paper deals with Markov decision processes with a countable state space. We demonstrate that a single, relatively simple condition suffices to guarantee that the value-iteration method converges and that an optimal policy can be computed via this method, once the existence of a solution to the average cost optimality equation has been established via any of the many available sets of existence conditions. © 1999 Elsevier Science B.V. All rights reserved.

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Many dynamic planning models can be formulated as a Markov decision process (MDP) with a *countable* state space and (possibly) *unbounded* costs. For example, in queueing and inventory models, one often encounters inventory levels or queue sizes and associated costs rising beyond any given level.

Let $S = \{0, 1, 2, \dots\}$ denote the state space of the MDP. For each state $i \in S$, the set of actions is given by $A(i)$. If in state $i \in S$, we choose $a \in A(i)$, a one-step expected cost $C(i, a)$ is incurred. At the beginning of the next period, the system is in state j with probability $P_{ij}(a)$. Let $v_n(i)$ be the minimum expected cost over a planning horizon of n periods, when starting in state i . The functions $\{v_n(i)\}$ satisfy the following recursion for all $n \geq 1$:

$$v_n(i) = \min_{a \in A(i)} \left\{ C(i, a) + \sum_{j \in S} P_{ij}(a) v_{n-1}(j) \right\}, \quad v_0(\cdot) = 0. \quad (1)$$

A fundamental question is how $v_n(\cdot)$ varies as the planning horizon, n , increases and how it relates to g^* , the long-run average cost value over an infinite planning horizon, assumed, as in most models, to be independent

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of the starting state. The key to the analysis is the optimality equation

$$v(i) = \min_{a \in A(i)} \left\{ C(i, a) - g^* + \sum_{j \in S} P_{ij}(a) v(j) \right\}, \quad i \in S. \quad (2)$$

In MDPs with finite state and action spaces, a solution to Eq. (2) always exists to which $\{v_n(\cdot) - ng^*\}$ converges, provided that some optimal policy is aperiodic, i.e., it induces an aperiodic Markov chain. Even if all optimal policies are periodic, all sequences of the type $\{v_{nJ+l} - (nJ+l)g^*\}_{n=1}^{\infty}$ converge to a solution of Eq. (2), where J is an identifiable integer related to the pure, stationary policies' periodicities, see [6] for details. Thus, when $\{v_n(\cdot) - ng^*\}$ converges, $\{v_n(\cdot)\}$ behaves asymptotically as an affine function with $(g^*, \dots, g^*)$ as the slope and a solution v^* to (2) as the intercept. Moreover, convergence enables a method for the computation of an optimal policy since it ensures that $\lim_{n \rightarrow \infty} g_{f_n} = g^*$, when f_n is any policy which is optimal with n periods to go (i.e., which achieves the minima in (1)) and g_{f_n} its long-run average cost. This method is referred to as *the value-iteration method*, and it is the most widely used general-purpose algorithm.

In this paper, we identify easily verifiable conditions under which these results carry over when the state space is countable. In this case, even the *existence* of an average-cost optimal policy or a solution of the optimality equation (2) is in question (see [15,16]). Most of the literature has focused on sufficient conditions for these existence questions, see e.g. [15,18].

In contrast, only a handful of papers have dealt with the use of general purpose methods, such as value iteration, for infinite state space models. Authors dealing with specific planning models have therefore felt compelled to develop problem specific exact solution methods, or analytical or simulation based heuristics, see e.g. Section 4.

In this paper we demonstrate that a *single*, relatively simple condition (C) suffices to guarantee that the value iteration method converges, once the existence of a solution to Eq. (2) has been verified with the help of *any* of the above mentioned sets of conditions. Moreover, some of the latter sets of conditions imply (C); for others we demonstrate that (C) can be established by the same arguments used to verify the existence conditions themselves.

Prior work on value iteration for infinite state space models is limited to the seminal papers by Hordijk et al. [12] for *bounded* one-step expected costs and Sennott [19] for the general case with (possibly) *unbounded* costs. Ref. [19] also identifies a condition which guarantees convergence of the value iteration method, provided *existence* of a solution to the optimality equation is established via the conditions in [18]. The new condition (C) has several advantages: first, it is usable in conjunction with *any* proof technique establishing the existence of a solution to the optimality equation, not just the conditions in [18]. Second, the proof of the validity of (C) is brief and elementary; third, it permits a rather general chain structure for stationary policies, and finally, we argue that (C) is more easily verified. Recently, Ghosh and Marcus [9] and Sennott [20] established conditions for a weaker convergence result of $\{v_n/n\}$.

1. Convergence of the value-iteration method

We assume that all one-step expected costs are uniformly bounded from below; i.e., $C(i, a) \geq -N$ for some constant N and all i, a . Without additional loss of generality, we assume $N = 0$ (i.e. $C \geq 0$); if $N > 0$, replace all $C(i, a)$ by $\bar{C}(i, a) = C(i, a) + N$ without affecting the set of optimal policies. The action sets are finite; the extension to compact metric action sets is rather straightforward under appropriate continuity assumptions with respect to a of $P_{ij}(a)$, $C(i, a)$ and certain expected functionals of the form $\sum_j P_{ij}(a)r(j)$.

Let Θ denote the class of all policies. A policy is called stationary if it prescribes the same (randomization of) action(s) $a \in A(i)$ whenever the system is in a given state $i \in S$, regardless of the system's prior history.

For any stationary policy f , let $P(f)$ denote the associated transition probability matrix, $P^n(f)$ its n th power, $\pi(f)$ the Cesaro-limit of $\{P^n(f)\}_{n=1}^{\infty}$, and $R(f)$ its set of recurrent states. Given a (possibly non-stationary) policy θ , let

$$g_{\theta}(i) \doteq \limsup_{N \rightarrow \infty} \frac{1}{N} \mathbf{E}_{\theta} \left[\sum_{n=0}^{N-1} C(X_n, a_n) \mid X_0 = i \right].$$

Thus, $g_{\theta}(i)$ denotes the expected average cost under θ when starting in state i , where X_n is the state and a_n the action chosen in period n . Let $g^*(i) = \inf_{\theta} g_{\theta}(i)$; a policy θ is *optimal* if $g_{\theta} = g^*$. We confine ourselves to the case where for some constant g^* , $g^*(i) = g^*$ for all $i \in S$. This condition prevails in almost all applications. We expect that our results can be extended to the most general case following the techniques in [6] and in [17] for multi-chain finite and infinite MDPs, respectively. The following is our basic assumption:

Assumption A. (i) *There exists a non-negative solution v^* to the optimality equation (2).* (ii) *No stationary policy f induces a Markov chain with null recurrent states.*

Assumption A is satisfied for all MDPs with finite state and action spaces, see [16]. In contrast, no stationary policy needs to be optimal when the state space is infinite (see [16]). However, as mentioned, several sets of sufficient conditions for the existence of an optimal stationary policy, and the somewhat stronger Assumption A, are known; in particular, the more recent sets of conditions are increasingly simple to verify, see e.g. [4]. Note, A (i) is equivalent to the existence of a solution v^* which is uniformly bounded from below: if v^* satisfies (1), so does $\{v^*(i) + c\}$ for any constant $c \geq 0$.

Our main objective is to show that if the state space is countable but Assumption A is satisfied, $\{v_n - ng^*\}_{n=1}^{\infty}$ (or the sequences $\{v_{nJ+l} - (nJ+l)g^*\}_{n=1}^{\infty}$ for an appropriate value of J and all $l = 0, \dots, J-1$) continue to converge to a solution v^* of Eq. (2) under a single condition B.

Condition B. (i) $B(i) \doteq \sup_n \mathbf{E}_{\theta}[v^*(X_n) \mid X_0 = i] < \infty$ for all $i \in S$ and any policy θ . (ii) For any stationary optimal policy f^* , $\mathbf{E}_{f^*}[\sum_{n=1}^T v^*(X_n) \mid X_0 = i] < \infty$, where $T = \min\{n \geq 1 : X_n \in R(f^*)\}$ is the first entry time into $R(f^*)$.

Theorem 1. Assume that A and B hold for a given solution v^* and stationary policy f^* . Then,

- (a) If f^* is aperiodic, $\{v_n - ng^*\}_{n=1}^{\infty}$ converges.
- (b) If f^* has periodicity $J \geq 2$, $\{v_{nJ+l} - (nJ+l)g^*\}_{n=1}^{\infty}$ converges for all $l = 0, \dots, J-1$.

Proof. (a) The proof is an extension of that in [12] to allow for unbounded one-step expected costs and hence unbounded value functions. With *bounded* one-step expected costs, the key to the proof in [12] is to demonstrate that the sequence $\{e_n \doteq v^* - (v_n - ng^*)\}$ is bounded in the regular l_{∞} -norm. Under Assumptions A and B, we show that the sequence $\{e_n\}$ is bounded in the $\|\cdot\|_B$ norm defined by $\|v\|_B \doteq \sup_{i \in S} (|v(i)|/B(i))$. As with the l_{∞} -norm, convergence in the $\|\cdot\|_B$ norm implies *pointwise* convergence. Note that

$$v^*(i) \leq C(i, f_n(i)) - g^* + \sum_j P_{ij}(f_n(i))v^*(j), \quad i \in S. \quad (3)$$

Subtracting

$$v_n(i) = C(i, f_n(i)) + \sum_j P_{ij}(f_n(i))v_{n-1}(j), \quad i \in S \quad (4)$$

from Eq. (3) and adding ng^* to both sides of the resulting inequality we obtain $e_n \leq \sum_j P_{ij}(f_n(i))e_{n-1}(j)$ for all $i \in S$. Iterating the inequality n times, we obtain

$$\begin{aligned} e_n(i) &\leq \sum_j P(f_n)P(f_{n-1}) \cdots P(f_1)_{ij} e_0(j) = \sum_j P(f_n)P(f_{n-1}) \cdots P(f_1)_{ij} v^*(j) \\ &= E_{\theta_n}[v^*(X_n) | X_0 = i] \leq B(i), \quad i \in S. \end{aligned} \quad (5)$$

Similarly, if f_n is replaced by f^* , Eq. (4) holds as an inequality and (3) holds as an equality. Repeating the above argument, we obtain

$$0 \leq \sum_j P_{ij}^n(f^*)e_0(j) \leq \sum_j P_{ij}(f^*)e_{n-1}(j) \leq e_n(i) \leq B(i), \quad n > 1, \quad (6)$$

completing the proof that for all n , e_n is non-negative and $\|e_n\|_B \leq 1$. Note that

$$\begin{aligned} e_{n+1}(i) &= v^*(i) + (n+1)g^* - v_{n+1}(i) = \max_{a \in A(i)} \left\{ b(i, a) + \sum_j P_{ij}(a)[v^*(j) + ng^* - v_n(j)] \right\} \\ &= \max_{a \in A(i)} \left\{ b(i, a) + \sum_j P_{ij}(a)e_n(j) \right\}, \end{aligned} \quad (7)$$

where $b(i, a) = v^*(i) + g^* - \sum_j P_{ij}(a)v^*(j) - C(i, a)$ and $\max_{a \in A(i)} b(i, a) = 0$ for all $i \in S$. Let $M(i) = \limsup_{n \rightarrow \infty} e_n(i)$ and $m(i) = \liminf_{n \rightarrow \infty} e_n(i)$. It follows from (6) that $0 \leq m(i) \leq M(i) \leq B(i)$ for all $i \in S$. As in [12] one verifies that

$$m(i) \geq \max_{a \in A(i)} \left\{ b(i, a) + \sum_j P_{ij}(a)m(j) \right\}, \quad (8)$$

$$M(i) \leq \max_{a \in A(i)} \left\{ b(i, a) + \sum_j P_{ij}(a)M(j) \right\}. \quad (9)$$

Let f be such that $f(i)$ maximizes the right-hand side of (9) for all $i \in S$. By Eqs. (8) and (9),

$$b(i, f(i)) + \sum_j P_{ij}(f(i))m(j) \leq m(i) \leq M(i) \leq b(i, f(i)) + \sum_j P_{ij}(f(i))M(j), \quad i \in S. \quad (10)$$

Multiply by $\pi_{ii}(f)$ and sum over $i \in S$ to obtain

$$\begin{aligned} \sum_i \pi_{ii}(f)M(i) &\leq \sum_i \pi_{ii}(f)b(i, f(i)) + \sum_i \pi_{ii}(f) \sum_{j=1}^{\infty} P_{ij}(f(i))M(j) \\ &= \sum_i \pi_{ii}(f)b(i, f(i)) + \sum_j M(j) \left\{ \sum_i \pi_{ii}(f)P_{ij}(f(i)) \right\} \\ &= \sum_i \pi_{ii}(f)b(i, f(i)) + \sum_j \pi_{jj}(f)M(j). \end{aligned} \quad (11)$$

(The second equality follows from $\pi = \pi P$, see e.g. [5].) To verify the first one, let $h_n(i) = \sum_{j=1}^n P_{ij}(f(i))M(j)$ and $h(i) = \sum_{j=1}^{\infty} P_{ij}(f(i))M(j) \leq \sum_{j=1}^{\infty} P_{ij}(f(i))B(j) < \infty$ by Condition B. Thus, by the Monotone Convergence Theorem, we obtain $\sum_i \pi_{ii}(f)h(i) = \sum_{i=1}^{\infty} \pi_{ii}(f) \lim_{n \rightarrow \infty} h_n(i) = \lim_{n \rightarrow \infty} \sum_{j=1}^n M(j) \{ \sum_{i=1}^{\infty} \pi_{ii}(f)P_{ij}(f(i)) \}$

$= \sum_{j=1}^{\infty} \pi_{jj}(f)M(j)$. Hence, $\sum_i \pi_{ii}(f)b(i, f(i)) \geq 0$, and since $b(\cdot, \cdot) \leq 0$, we have by (A) that $b(i, f(i)) = 0$ for all states $i \in R(f)$. Define f^* as a policy satisfying the optimality equation for $v = v^*$, choosing $f^*(i) = f(i)$ for all $i \in R(f)$. Clearly, $R(f) \subseteq R(f^*)$ and for $i \in R(f^*)$, $M(i) = m(i)$ follows from Lemma 3 in [19] (itself an adaptation of [12] to allow for unbounded value functions). To show the same for $i \notin R(f)$, note from Eq. (10) that for all $i \in S$: $0 \leq M(i) - m(i) \leq \sum_j P_{ij}(f(i))[M(j) - m(j)]$. Iterating this inequality N times and averaging over N we obtain, with $T = \min\{n: X_n \in R(f) | X_0 = i\}$ the first entry time into $R(f)$, that: $0 \leq M(i) - m(i) \leq \frac{1}{N} \sum_{l=1}^N P_{ij}^l(f)[M(j) - m(j)] \leq \frac{1}{N} \mathbf{E}_f \left\{ \sum_{l=1}^{\min(T, N)} [M(X_l) - m(X_l)] + \sum_{l=\min(T, N)+1}^N [M(X_l) - m(X_l)] | X_0 = i \right\} \leq \frac{1}{N} \mathbf{E}_f \left[\sum_{l=1}^T B(X_l) | X_0 = i \right]$, which by (B) converges to zero as $N \rightarrow \infty$. Thus, $\{e(n)\}_{n=1}^{\infty}$ converges and hence $\hat{v} = \lim_{n \rightarrow \infty} [v_n(i) - ng^*]$ exists.

To show that $\hat{v}$ satisfies the optimality equation (2), note that $v_n(i) - ng^* = \min_{a \in A(i)} \{C(i, a) - g^* + \sum_j P_{ij}(a)[v_{n-1}(j) - (n-1)g^*]\}$, take the limit as $n \rightarrow \infty$ on both sides, and interchange the “lim” and “min” operators since the function $h: \mathbf{R}^{|A(i)|} \rightarrow \mathbf{R} : (x_1, \dots, x_{|A(i)|}) \rightarrow \min_{a \in A(i)} x_a$ is continuous.

For part (b), by A (ii), $J < \infty$. Consider J -step transitions as a “super-step” transition and observe that $P^J(f^*)$ is aperiodic. The remainder of the proof is analogous to that of part (a). $\square$

Remark 1. The norm $\|\cdot\|_B$ under which the sequence $\{e_n\}$ is bounded is similar to that used for other analyses of MDPs with a countable state space, see e.g. [13], and also [7,17].

Remark 2. If the periodicity of the optimizing policy f^* is unknown, one can under A and B guarantee the convergence of $\{v_n - ng^*\}$ by applying the transformation: $\tilde{P}_{ij}(a) = \tau P_{ij}(a)$ for $i \neq j$ and $\tilde{P}_{ii}(a) = \tau P_{ii}(a) + 1 - \tau$ with $0 < \tau < 1$. After this transformation all stationary policies are aperiodic while the set of solutions v^* and optimizing policies f^* in (2) remain unaltered, see [6].

2. Implementation issues

Several implementation issues need to be considered. First, the infinite sums to the right of (1) can not be evaluated on a term by term basis, nor can the function $v_n(\cdot)$ be evaluated and stored for infinitely many values. Instead, the v_n -functions are evaluated in I states (for some large value of I) and the values $\{v_{n-1}(j), j > I\}$ approximated by a (polynomial) extrapolation of $\{v_{n-1}(j), j = 1, \dots, I\}$. See [14] for a review of other implementations of this type of approximation. This procedure is significantly faster and easier than approximating the infinite state space model by a series of truncated finite state space models. Second, the theory merely guarantees that $\lim_{n \rightarrow \infty} g_{f_n} = g^*$. In practice, one often observes the same policy f^* achieving the minima in (1) in all but a few initial, or at least finitely many, iterations. Such a policy optimizes average costs, see Theorem 3 of [19] and the remark, p. 304 of [12]. Also, as in [6], one easily proves that $g_{f_n} - g^* \leq \text{sp}[v_n - v_{n-1}]$, where $\text{sp}[x] = \max_i x(i) - \min_i x(i)$. Thus, the algorithm can be stopped when $\text{sp}[v_n - v_{n-1}]$ is sufficiently small. Third, instead of the functions v_n , which grow linearly with n , one should generate the normalized and bounded functions w_n , defined by $w_n(i) = v_n(i) - v_n(i_0)$ for some reference state $i_0 \in S$. See [6] for a recursion generating $\{w_n(\cdot)\}$ and the *same* sequence of policies $\{f_n\}$ as (1).

3. Sufficient conditions for B

Condition B is by itself difficult to verify since v^* , a solution to (2), is generally unknown. We first replace B by an equally simple, but in addition easily verifiable condition C.

In general, when the one-step expected costs $C(i, a)$ are unbounded, v^* is unbounded as well. However, we assume that an upper bound for the “growth” of v^* is known; that is, there exists a known function $r(i) \geq 1$

and a (possibly unknown) constant $K > 0$ such that

$$v^*(i) \leq Kr(i), \quad i \in S. \quad (12)$$

We refer to $r(\cdot)$ as an *order function* of v^* . Often, one can show that if the one-step costs $C(i, a)$ are bounded by a polynomial in i (i.e. $C(i, a) = O(|i|^\rho)$ for some $\rho \geq 1$, $i \rightarrow \infty$), a solution v^* of Eq. (2) exists and is bounded by a polynomial of a higher degree (e.g. $v^*(i) = O(|i|^{\rho+1})$ i.e., $r(i) = |i|^{\rho+1}$ is an order function of v^*). The knowledge of an order function $r(\cdot)$ allows us to replace B by

Condition C. Let $r(\cdot)$ be an order function of v^* . (i) There exists a class of policies $\Theta' \subseteq \Theta$ containing the finite and infinite horizon optimal policies, and a constant c such that for all $\theta \in \Theta'$, $E_\theta[r(X_n) | X_0 = i] \leq cr(i)$ for all $n \geq 1$ and $i \in S$. (ii) Also, B(ii) holds with v^* replaced by $r(i)$.

For example, if $r(i) = |i|^m$, Condition C consists of the verification that for $X_0 = i$ the m th moment of the system state at any point in time is $O(|i|^m)$.

Corollary 1. Theorem 1 holds with Condition B replaced by C.

Proof. The proof of Theorem 1 can be repeated, merely replacing the bound $e_n(i) \leq B(i)$ in (5) by $e_n(i) \leq Kcr(i)$. The latter follows since by the first inequality in (5), $e_n(i) \leq \sum_j P(f_n)P(f_{n-1}) \cdots P(f_1)_{ij} v^*(j) \leq K \sum_j P(f_n)P(f_{n-1}) \cdots P(f_1)_{ij} r(j) = K E_{\theta_n}[r(X_n) | X_0 = i] \leq Kcr(i)$, for all $i \in S$. $\square$

Remark 3. The conditions for convergence of $\{v_n - ng^*\}$ in Sennott [19] are as follows:

Assume Assumptions 1, 2 and 3* of [18] or [19] are satisfied, as well as:

Condition S. There exists a stationary policy f which induces an irreducible, aperiodic, and positive recurrent Markov chain with steady-state distribution $\pi(f)$ such that

- (i) $\limsup_{n \rightarrow \infty} E_{\theta_n}[r(X_n) | X_0 = i] \leq \limsup_{n \rightarrow \infty} E_f[r(X_n) | X_0 = i]$; $i \in S$,
- (ii) $\sum_i \pi(f)_i r(i) < \infty$.

(Note, Assumptions 1, 2 and 3* in [18] are sufficient for (A).) As stated above, we believe that Condition C here, has several advantages: First it is usable in conjunction with any proof technique establishing Assumption A, not just Assumptions 1, 2 and 3* of [19]. Second, C is often more easily verified since it is sufficient to show that the function $r(\cdot)$ prevails as an *order function* for the functions $\{E[r(X_n) | X_0 = \cdot]; n \geq 1\}$ rather than a precise bound; also, no information regarding the growth-rate of the steady-state distribution of any of the stationary policies is (explicitly) required. Third, C allows for a rather general chain structure; in practice, it permits some or all stationary policies to have transient states. Finally, it is clear from the proof of Theorem 1 that the essential part in extending the convergence result of [12] to models with *unbounded* value functions, consists in establishing that the sequence $\{e_n\}$ remains bounded in the $\|\cdot\|_r$ -norm. Condition C allows for a brief and self-contained condition for the latter.

We now identify several conditions under which C is guaranteed to hold. In particular,

C1. There exists a finite subset G of the state-space S and *some* order function r of v^* such that

$$\bar{r}(i) \doteq \max_{a \in A(i)} \left\{ \sum_j P_{ij}(a) r(j) \right\} < \infty, \quad i \in G, \quad (13)$$

$$\sum_j P_{ij}(a) r(j) \leq r(i), \quad i \notin G \text{ and all } a \in A(i). \quad (14)$$

C2. There exists a finite subset G of the state-space S and *some* order function r of v^* such that

$$\sum_{j \notin G} P_{ij}(a)r(j) \leq r(i), \quad \text{for all } i \in S \text{ and all } a \in A(i). \quad (15)$$

Theorem 2. $C1 \Leftrightarrow C2 \Rightarrow C$.

Proof. $C1 \Rightarrow C2$: Define the function r^* as follows: for $i \notin G$ set $r^*(i) := r(i)$, and for $i \in G$, $r^*(i) = \bar{r}(i)$. For $i \in G$, $r^*(i) \geq \sum_{j \notin G} P_{ij}(a)r^*(j)$ for all $a \in A(i)$. For $i \notin G$, $r^*(i) = r(i) \geq \sum_{j \notin G} P_{ij}(a)r(j) = \sum_{j \notin G} P_{ij}(a)r^*(j)$. In other words, the function r^* satisfies Eq. (15) and it is an order function of v^* since it modifies the order function r in only finitely many states.

$C2 \Rightarrow C1$: Define the function r^* as follows: for all $i \in G$, $r^*(i) := 1$; otherwise, $r^*(i) := r(i) + 1$. For $i \in G$ and all $a \in A(i)$, $\sum_j P_{ij}(a)r^*(j) = \sum_{j \in G} P_{ij}(a) + \sum_{j \notin G} P_{ij}(a)[r(j) + 1] = 1 + \sum_{j \notin G} P_{ij}(a)r(j) \leq 1 + r(i) < \infty$, verifying (13). Also, for $i \notin G$ and all $a \in A(i)$, $\sum_j P_{ij}(a)r^*(j) = \sum_{j \in G} P_{ij}(a) + \sum_{j \notin G} P_{ij}(a)[r(j) + 1] = 1 + \sum_{j \notin G} P_{ij}(a)r(j) \leq 1 + r(i) = r^*(i)$, verifying Eq. (14). Finally, r^* is an order function for v^* since $r^*(i) \geq r(i)$ for all but finitely many values of i .

$C2 \Rightarrow C$: Fix a policy $\theta \in \Theta'$ and a starting state $X_0 = i \in S$. Let τ_l denote the time of the l th visit to set G , i.e.: $\tau_0 \doteq 0$, $\tau_l \doteq \min\{n > \tau_{l-1} : X_n \in G\}$. For any event E , let $\mathbf{1}\{E\} = 1$ if the event occurs, and $\mathbf{1}\{E\} = 0$ otherwise. Note that

$$\mathbf{E}_\theta[r(X_n) | X_0 = i] = \mathbf{E}_\theta[r(X_n)\mathbf{1}\{\tau_1 > n\} | X_0 = i] + \sum_{l=1}^{\infty} \mathbf{E}_\theta[r(X_n)\mathbf{1}\{\tau_l \leq n < \tau_{l+1}\} | X_0 = i]. \quad (16)$$

The first term in Eq. (16) is bounded by $r(i)$ since by repeated use of Eq. (15): $r(i) \geq \mathbf{E}_\theta[r(X_n)\mathbf{1}\{\tau_1 > n\} | X_0 = i]$. The second term in Eq. (16) equals $\sum_{l=1}^{\infty} \Pr\{\tau_l \leq n < \tau_{l+1}\} \mathbf{E}_\theta[r(X_n) | X_0 = i, \tau_l \leq n < \tau_{l+1}]$ and by the Markov property of the process $\{X_n\}$,

$$\begin{aligned} \mathbf{E}_\theta[r(X_n) | X_0 = i, \tau_l \leq n < \tau_{l+1}] &\leq \max_{j \in G} \mathbf{E}_\theta[r(X_n) | X_0 = i, X_{\tau_l} = j, \tau_l \leq n < \tau_{l+1}] \\ &\leq \max_{j \in G} \sup_{p \geq 0} \mathbf{E}_\theta[r(X_p)\mathbf{1}\{\tau_1 > p\} | X_0 = j] \leq \max_{j \in G} r(j) \doteq R. \end{aligned}$$

Thus, $\mathbf{E}_\theta[r(X_n) | X_0 = i] \leq r(i) + R \leq (1 + R)r(i)$ for all $i \in S$. $\square$

Conditions C1 and C2 are related to the existence of a $C(\cdot, \cdot)$ -excessive function, a concept introduced in [12], to part (i) of the central *Lyapunov* condition in [17] and to Lippman [13]'s condition for bounding functions in discounted models. C1 is a weaker version of the condition in [19], shown to be sufficient to guarantee part (ii) of her value-iteration convergence Condition S, see Remark 3.

Theorems 1 and 2 show in fact that C1 by itself guarantees convergence of the value-iteration method whenever Assumption A applies. Alternatively, let $v_\alpha^*(i)$ denote the minimal expected system-wide costs when future costs are discounted with a discount factor $\alpha < 1$.

Corollary 2. Assume Condition C1 (or C2) holds for some arbitrary finite function $r(i)$ with $r(i) \geq 1$. Assume in addition that

(i) For some $\varepsilon > 0$, $C(i, a) \geq \varepsilon$, for all $i \notin G$ and all $a \in A(i)$. i.e., the one-step expected costs are bounded away from zero in every state $i \notin G$.

(ii) There exists a stationary policy f inducing an irreducible, aperiodic and positive recurrent Markov chain.

(iii) There exists a state $i_0 \in S$ and a constant $N < \infty$ such that $v_\alpha^*(i) - v_\alpha^*(i_0) \geq -N$ for all $i \in S$ and all $\alpha \in (0, 1)$.

Then, $\{v_n(\cdot) - ng^*\}$ converges to a solution v^* of the optimality equation (2).

Proof. By Proposition 1 of [19], we have that (C1) in conjunction with (i) and (ii) apply Assumptions 1 and 3* in [18]. Since condition (iii) is equivalent to Assumption 2, *ibid*, it follows from the Theorem in [18] that Assumption A applies. The corollary now follows from Theorem 1 and Theorem 2. $\square$

Finally, in many applications, it is relatively easy to show that for an appropriately chosen finite subset G of the state space S and an appropriate subset of policies $\Theta' \in \Theta$ which includes the optimal finite and infinite horizon policies, the maximum expected cost and time until reaching the set G is finite for every starting state $i \in S$, i.e.,

Condition C3.

$$C_G(i) \doteq \sup_{\theta \in \Theta'} E_\theta \left[\sum_{n=0}^{\tau} \max(1, C(X_n, a_n)) | X_0 = i \right] < \infty; \quad i \in S,$$

where $\tau = \min \{n \geq 1 | X_n \in G, X_0 = i\}$ is the first visit time to the set G . Combined with a technical regularity condition stated below, Condition C3 guarantees *simultaneously* Assumption A as well as Condition C:

C3(i): Let $G^* = \{i \in G : i \text{ is positive recurrent under some stationary policy } f\}$. Then, any state $j \in G^*$ can be reached from any state $i \in G^*$ under some policy $\theta \in \Theta'$.

Corollary 3. Assume Conditions C3 and C3(i) hold,

(a) There exists a solution v^* of the optimality Eq. (2) and $r(\cdot) \doteq C_G(\cdot)$ is an order function of v^* . Let f^* be a policy which achieves the minima in (2) for $v = v^*$.

(b) If f^* is aperiodic, $\{v_n - ng^*\}_{n=1}^\infty$ converges.

(c) If f^* is periodic with periodicity J , $\{v_{nJ+l} - (nJ+l)g^*\}_{n=1}^\infty$ converges for all $l = 0, \dots, J-1$.

Proof. Part (a): Immediate from Theorem 2 in [7]. Note that Assumption 3 there holds since the one-step expected costs are bounded from below. Moreover, Conditions C3 and C3(i) correspond with Assumptions 2(a) and 2(b). Assumptions 1 and 4 are trivially satisfied with finite action sets.

Parts (b) and (c): In view of Theorems 1 and 2, it suffices to verify that Condition C2 holds. Let $r \doteq C_G$. It follows from part (a) that r is an order function of v^* . Since $r = C_G$ represents a maximum expected total reward function, with non-negative one-step expected “rewards”, it follows from (case N of) Proposition 9.8 in Bertsekas and Shreve [3] that

$$r(i) = \sup_{a \in A(i)} \left\{ \max[1, C(i, a)] + \sum_{j \notin G} P_{ij}(a) r(j) \right\} \geq \max[1, C(i, a)] + \sum_{j \notin G} P_{ij}(a) r(j)$$

for all $i \in S$ and all $a \in A(i)$; i.e., (C2) holds. $\square$

In some models, it is easiest to verify B or C directly; in others, it is easier to employ one of the sufficient conditions C1, C2 or C3.

Example. Consider the following discrete-time queueing model. Customers arrive to a single server. There is a probability $p_j > 0$ of j ($j \geq 0$) customers arriving in any period. $\lambda = \sum_j j p_j < \infty$. Arrivals are independent of service times which are i.i.d. and geometrically distributed with rate $a\beta \leq 1$, where a , the action, can be

chosen, in each period, from a finite set. With i , the number of customers at the beginning of a period, we have $C(i, a) = Q(i) + B(a)$ with $Q(\cdot)$ a polynomial of degree n . Using standard arguments, e.g. the conditions in [18], one verifies that (A) holds with $r(i) = i^{n+1}$, an order function of v^* . It follows from Eq. (14) in [18] that for all a , $\sum_j P_{ij}(a)r(j) - r(i)$ is a polynomial of degree n , with negative leading coefficient, thus verifying C1. C1 or C2 arise easily verifiable conditions in a similar queueing model with controllable arrival rates, see [19]. We now turn to an example in which B is verified directly.

4. The single item inventory model with limited production capacity: an example

Consider the following single item inventory model: Inventory is monitored periodically. Demands in consecutive periods are i.i.d., as the integer-valued random variable D . Let $p_j = \Pr\{D = j\}$, $j \geq 0$. Replenishment orders at the beginning of each period are subject to a production capacity of b units. Stockouts are backordered. Production costs are proportional to the production volume. Their long-run average value is the same under all strategies resulting in a stable system and can thus be ignored. There are holding and backlogging costs which are accounted for by charging a cost $L(y)$ in any period in which the inventory position (= inventory on hand + orders outstanding – backlogs) after ordering equals y , with $L(\cdot)$ an arbitrary convex function. For example, when each unit carried in stock (backlogged) costs h (p) per period, and orders arrive after a fixed lead time of λ periods, $L(y) = hE[y - (D_1 + \dots + D_\lambda)]^+ + pE[(D_1 + \dots + D_\lambda) - y]^+$, where $D_1, D_2, \dots$ are i.i.d. as D .

For this model, Tayur [22] developed an exact solution method, Glasserman and Tayur [11] a simulation-based heuristic, and Glasserman [10] approximations and bounds for the optimal policy performance. All of these methods are tailored to this specific model and exploit its special characteristics ([10,11] deal in addition with a multi-echelon version of this model).

We now show that the value-iteration method can be used as a simple alternative. This reduces to the recursion:

$$v_n(x) = \min_{x \leq y \leq x+b} \left\{ L(y) + \sum_{j=0}^{\infty} p_j v_{n-1}(y-j) \right\}, \quad n \geq 1; \quad v_0 = 0; \quad x \in S \quad (17)$$

converging to a solution v^* of Eq. (2) under the *same* minor regularity conditions used to prove the *existence* of a solution to Eq. (2). (Note, S contains *all* integers.) These conditions are:

Assumption A1. (a) $\lim_{|y| \rightarrow \infty} L(y) = \infty$; (b) L is non-negative and convex and (c) $L(y) = O(|y|^\rho)$ as $|y| \rightarrow \infty$ for some $\rho > 0$, i.e. $L(y) \leq c|y|^\rho$ for some $c > 0$.

Assumption A2. $E[D^{\rho+2}] < \infty$,

Assumption A3. $b > E[D]$.

A1 is straightforward under most common cost and lead-time structures, e.g., the example above, where it applies with $\rho = 1$ and A2 reduces to D having a finite variance. A3 is the necessary and sufficient condition for the system's stability; i.e., for the production facility's utilization rate to be less than one.

It is easily verified in view of A1 and A2 that all value functions $v_n(\cdot)$ are convex and that $\lim_{|x| \rightarrow \infty} v_n(x) = \infty$. This observation permits a significant simplification in the implementation of the value-iteration method. At the n th iteration, it suffices to determine the (smallest) minimum y_n of the minimand $[L(y) +$

$\sum_j p_j v_{n-1}(y-j)$]; with y_n determined, any required function value can be evaluated as follows:

$$v_n(x) = \begin{cases} L(x) + \sum_j p_j v_{n-1}(x-j), & x \geq y_n, \\ L(y_n) + \sum_j p_j v_{n-1}(y_n-j), & y_n - b \leq x < y_n, \\ L(x+b) + \sum_j p_j v_{n-1}(x+b-j), & x < y_n - b, \end{cases}$$

i.e., the inventory position is always increased to a level as close as possible to y_n , a so-called *modified base-stock policy*. (If L is a polynomial of degree q , we suggest approximating the tail of the functions $v_n(\cdot)$ by a polynomial of degree $q+1$).

Ref. [8] proved that A applies under regularity conditions, somewhat stronger than A1–A3. Using more recent conditions for A, Aviv and Federgruen [2] recently provided a simpler proof based on A1–A3 only. (Their paper deals with an extension of the model in which all parameters vary according to a periodic pattern.)

Corollary 4. Assume A1–A3 hold. Then $\{v_n - ng^*\}$ converges to a solution v^* of (2).

Proof. Theorem 4 in [2] shows that a non-negative solution v^* exists for the optimality equation (2) with $r(x) = |x|^{\rho+1} + 1$ as an order function of v^* . Moreover, Lemma 3 in [2] shows that a pair of inventory levels $\bar{u} \geq 0$ and $\underline{l} \leq 0$ exist such that in states $x \geq \bar{u}$ ($x \leq \underline{l}$) only zero (full capacity) orders need to be considered, without loss of optimality. Under a given policy (which satisfies this restriction) let $\{X_n\}$ and $\{D_n\}$ denote the sequences of states and demands observed, with $X_0 = x$. Observe that

$$\min\{X_n, \underline{l}\} \leq X_n \leq \max\{x, \bar{u} + b\} \quad \text{for all } n \geq 1. \quad (18)$$

(Let $n_0 = \min\{n \geq 0: X_n \leq \bar{u}\}$; X_n is non-decreasing during the first n_0 periods and thereafter the inventory level after ordering is never in excess of $\bar{u} + b$, thus verifying the upper bound in Eq. (18).) By the convexity of the function $|x|^{\rho+1}$, we obtain:

$$\begin{aligned} |X_n|^{\rho+1} &\leq \max\{(-\min\{X_n, \underline{l}\})^{\rho+1}, (\max\{x, \bar{u} + b\})^{\rho+1}\} \\ &\leq (-\min\{X_n, \underline{l}\})^{\rho+1} + \max\{|x|^{\rho+1}, (\bar{u} + b)^{\rho+1}\}. \end{aligned} \quad (19)$$

Thus, in order to show C(i), i.e. $E[r(X_n)|X_0 = x] = O(r(x))$, it suffices to show that $E[(-\min\{X_n, \underline{l}\})^{\rho+1}|X_0 = x] = O(|x|^{\rho+1})$. To do so, define a positive drift *reflected random walk* as follows: $R_0 = \min\{x, \underline{l}\}$; $R_{n+1} = \min\{R_n + b - D_n, \underline{l}\}$, with increments distributed as $I \doteq (b - D)$. We show by induction, that $R_n \leq \min\{X_n, \underline{l}\} + b$: observe first that $R_0 = \min\{x, \underline{l}\} \leq \min\{X_0, \underline{l}\} + b$. Now assume that the bound holds for some $n \geq 0$. Consider the following two cases:

$$\begin{aligned} X_n > \underline{l}: R_{n+1} &= \min\{R_n + b - D_n, \underline{l}\} \leq \min\{X_n + b - D_n, \underline{l}\} \\ &\leq \min\{X_n - D_n, \underline{l}\} + b \leq \min\{X_{n+1}, \underline{l}\} + b, \end{aligned}$$

$$\begin{aligned} X_n \leq \underline{l}: R_{n+1} &= \min\{R_n + b - D_n, \underline{l}\} \leq \min\{\min\{X_n, \underline{l}\} + b + b - D_n, \underline{l}\} \\ &= \min\{X_n + b - D_n + b, \underline{l}\} = \min\{X_{n+1} + b, \underline{l}\} \leq \min\{X_{n+1}, \underline{l}\} + b, \end{aligned}$$

thus concluding that $(-\min\{X_n, \underline{l}\})^{\rho+1} \leq (|R_n| + b)^{\rho+1}$.

To show that $E[(-\min\{X_n, l\})^{\rho+1} | X_0 = x] = O(|x|^{\rho+1})$, it thus suffices to verify that $E[|R_n|^{\rho+1} | X_0 = x] = O(|x|^{\rho+1})$. Now, for all $n \geq 1$, $R_n \doteq \min\{x, l\} + \min\{S_n, S_n - S_1, S_n - S_2, \dots, S_n - S_{n-1}, 0\}$, where S_n denotes the sum of n independent increments all distributed as I , see Asmussen ([1], Proposition 1.1 on p. 181). The sequence $\{R_n\}$ is thus stochastically decreasing with a limiting distribution equal to that of $[\min\{x, l\} + R^*]$ where $E[|R^*|^{\rho+1}] < \infty$ since $E[|I|^{\rho+2}] < \infty$, see Theorem 2.1 on p. 184 of [1]. We conclude that a constant C exists such that for all $n \geq 1$ $E[|R_n|^{\rho+1} | X_0 = x] \leq C(|x|^{\rho+1} + 1)$. C(ii) is easily verified, see e.g. Proposition 1 in [2] or Theorem 1 in [8], replacing the one-step cost function C by r , and using A2. $\square$

This illustrates our claim that in general the marginal effort to verify the single Condition C, beyond what is required to verify Assumption A, is slight.

Finally, as mentioned in [2,8], a stationary policy continues to be optimal if the order costs have an additional *fixed* component. The conditions used to establish A continue to apply, as is straightforwardly verified. On the other hand, under this generalized cost structure, Shaoxiang and Lambrecht [21] showed that the optimal policy fails to be of a simple structure. Fortunately, C can be demonstrated in the same way as in the basic model (without fixed order costs). Thus, value-iteration can again be used to identify an optimal policy (see Remark 2); moreover, in this generalized model it appears to be the only known solution method.

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